SECTION 1 — MODEL DESCRIPTION

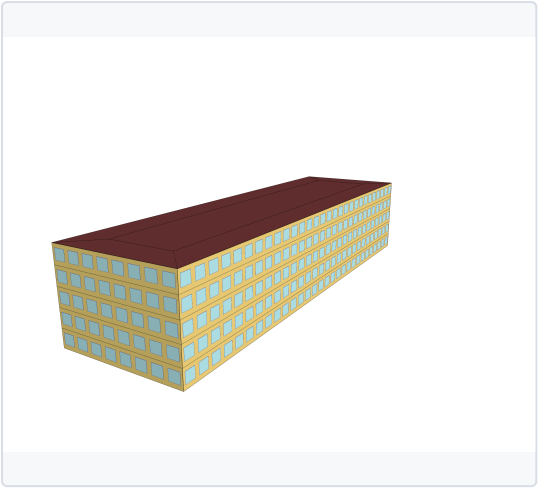
BUILDING DESCRIPTION

Sector	Commercial-Institutional
Building type	Office
Building subtype	Mid-rise (4-9 storeys)
Vintage	After 2010

HVAC SYSTEM

Heating	Electric boiler (hot water) + Hydronic baseboards / radiators
Cooling	Water-cooled chiller
Air distribution	Constant air volume (CAV) (1 air loop)
Heat recovery	None

3D MODEL



Three-dimensional geometric representation of the building model used for energy simulation.

BUILDING CHARACTERISTICS

Total floor area [m²]	10,738
Conditioned floor area [m²]	10,738
Above-ground storeys	6
Climate zone	7
Weather file	Montreal-Trudeau Intl AP QC CAN CWEC2016 WMO#=716270

WINDOW-TO-WALL RATIO [%]

Average	40.0
North	40.0
East	40.0
South	40.0
West	40.0

ENVELOPE & INTERNAL LOADS

INSULATION & AIR TIGHTNESS

Exterior walls [W/m²K]	0.20
Roof [W/m²K]	0.21
Exposed floors / slabs [W/m²K]	2.95
Infiltration [m³/h/m² ext. surface]	0.82

GLAZING

Glazing U-factor [W/m²K]	2.50
SHGC	0.70

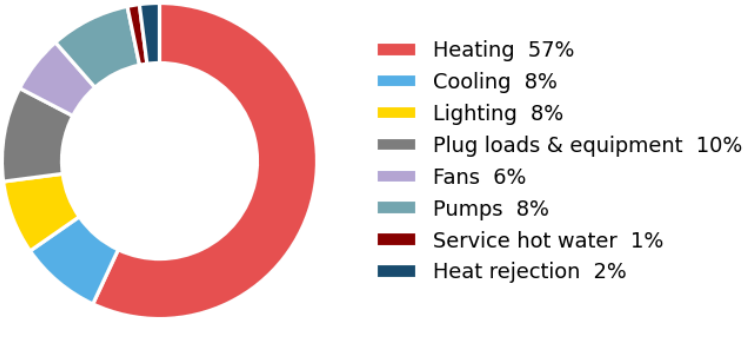
INTERNAL LOADS (MODEL INPUTS)

Occupant density [m²/person]	14.0
Lighting power density [W/m²]	6.7
Equipment power density [W/m²]	7.5

ANNUAL END USES [kWh/year]

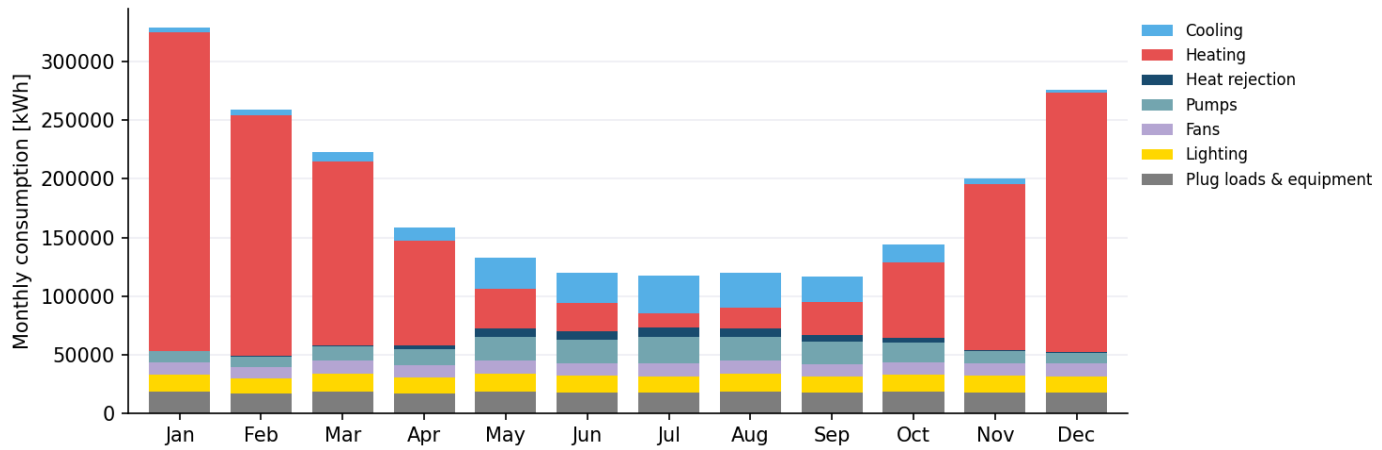
END USE	ELECTRICITY	DISTRICT HEATING	TOTAL	kWh/m²
Heating	1,266,817	0	1,266,817	118.0
Cooling	186,725	0	186,725	17.4
Interior lighting	168,292	0	168,292	15.7
Plug loads & equipment	215,817	0	215,817	20.1
Fans	131,158	0	131,158	12.2
Pumps	182,553	0	182,553	17.0
Heat rejection	45,364	0	45,364	4.2
Service hot water	0	27,569	27,569	2.6
Total			**2,224,297**	**207.1**

SHARE OF ANNUAL CONSUMPTION



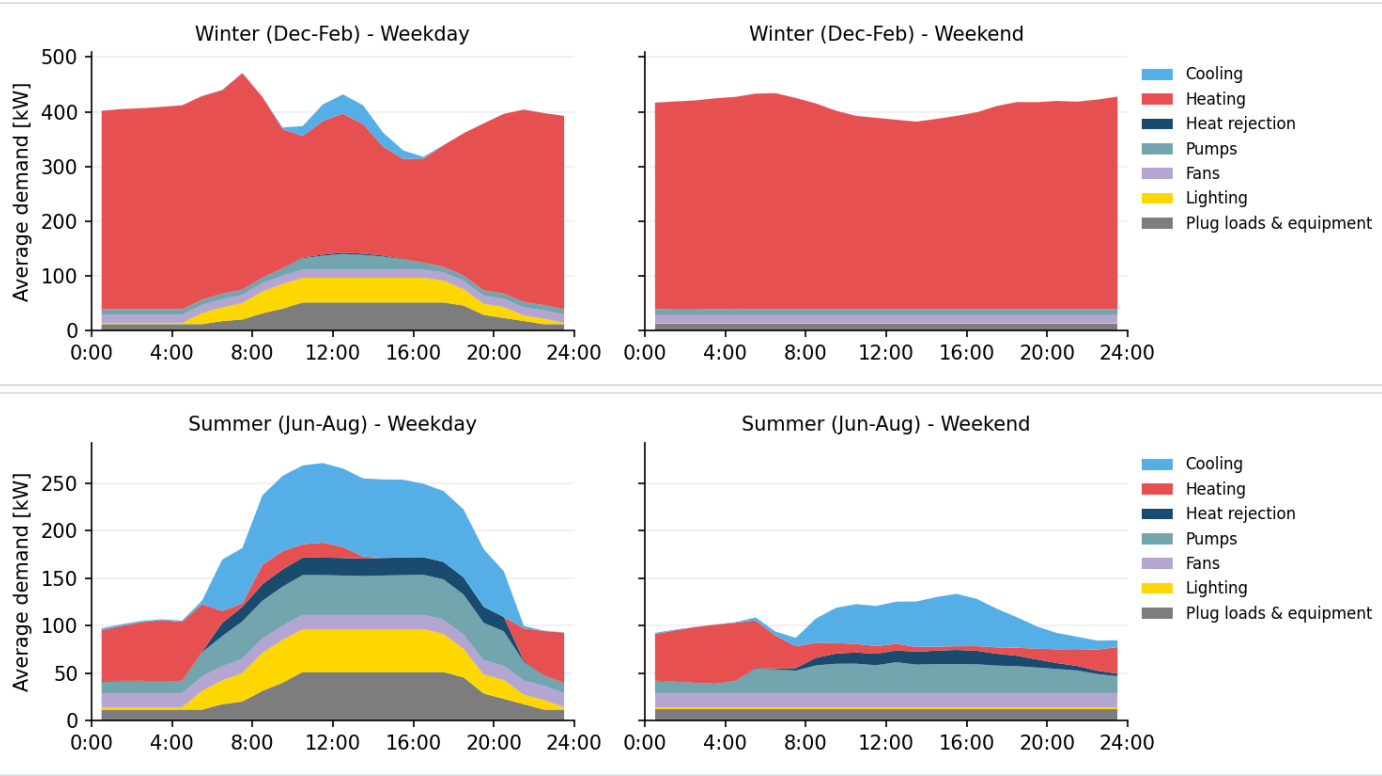
ENERGY USE INTENSITY	TOTAL CONSUMPTION	ELECTRICITY	NATURAL GAS	OTHER FUELS	PEAK ELECTRIC DEMAND
207 kWh/m²/yr (total area)	2,224,297 kWh/yr	2,196,725 kWh/yr	0 kWh/yr	27,569 kWh/yr	1,083 kW · 100.8 W/m²

MONTHLY CONSUMPTION BY END USE



Monthly energy consumption by end use, all fuels combined.

TYPICAL DAY PROFILES — WEEKDAY VS. WEEKEND



Average hourly demand by end use for winter and summer. Weekdays (Monday to Friday) are shown on the left and weekends and holidays on the right, showing the office occupancy schedule and HVAC set-back.

NOTES & ASSUMPTIONS

- Archetype energy model simulated with OpenStudio/EnergyPlus. Values are model outputs for the reference weather year and are not measured data.
- Energy use intensity is normalized by the total floor area (10,738 m²). Offices are not normalized per unit.
- Archetype: Mid-rise office (4-9 storeys), built after 2010.
- HVAC entries marked "(typical)" are not modelled explicitly and come from the archetype assumption table.
- Peak electric demand occurs on 24-DEC-05:10.